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Motivations for motorcycle use for Urban travel in Latin America: A qualitative study

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ABSTRACT

Motorcycle use for utilitarian trips in Latin American cities has grown significantly in recent years. The researchers used qualitative methods to understand the motivations of motorcycle users that might contribute to this growth in six cities: Barranquilla, Bogotá (Colombia), São Paulo, Recife (Brazil), Caracas (Venezuela), and Buenos Aires (Argentina). Researchers used semi-structured interviews and focus groups to gather data from six categories of motorcycle users: motorcycle taxi drivers, motorcycle taxi users, motorcyclists for delivery, motorcyclists for private use, owners in the process of selling their motorcycles, and potential motorcyclists (those seeking to buy motorcycles). Common themes emerged across the six cities, including the time advantage that motorcycles offered versus deficient public transportation and congested auto traffic, the low cost of motorcycles versus other transport modes, the vulnerability of motorcyclists to traffic injury and death, and cultural aspects of motorcycle use. Policy implications include the need to make motorcycle travel safer and improve public transportation in Latin American cities.

1. Introduction

Motorcycle use has grown dramatically worldwide in recent decades (Haworth, 2012), with considerable increases in Latin American cities. This mode provides large mobility gains for users but also has important drawbacks, including elevated traffic casualties. Despite the increased health risks associated with motorcycles, they are increasingly popular in the region.

The researchers gathered data from motorcycle users in six Latin American cities: Bogotá and Barranquilla (Colombia), São Paulo and Recife (Brazil), Caracas (Venezuela), and Buenos Aires (Argentina) (Fig. 1). This includes South America's largest city (São Paulo), three capital cities that are the largest in their respective countries (Bogotá, Caracas, Buenos Aires), and two state capitals (Barranquilla, Recife). The data was collected for a qualitative study on motorcycle users' perceptions of this mode that the authors completed for the Corporación Andina de Fomento (CAF) – Development Bank of Latin America (Pardo et al., 2012), and that is included in a book on the topic (Rodríguez et al., 2015).[1]

Instead of addressing a specific hypothesis, the authors set out to achieve the broad objective of developing a better understanding of the motorcycle user perceptions fueling the growing use of motorcycles in six Latin American cities. To achieve this goal, researchers conducted interviews and focus groups with motorcycle users and potential users. In the analysis, the authors employed a 'bottom-up' approach, where the themes of analysis emerged from the data collected.

2. Literature review

While leisure use of motorcycles predominates in developed countries, use for utilitarian trips is more prevalent in developing countries (Rogers, 2008). Recent increases in ownership rates for motorcycles, as for all motorized vehicles, are particularly pronounced in emerging markets (Dargay et al., 2007). Although most growth has occurred in Asia, motorcycles have become increasingly ubiquitous in Latin American countries. For example, while auto sales in Brazil increased four times between 1992 and 2007, motorcycle sales increased by a factor of 12 during the same period (Eduardo A. Vasconcellos, 2008). The Brazilian city of São Paulo witnessed an increase from a fleet of 50,000 motorcycles in 1990 to 500,000 in 2007. In Colombia, motorcycle sales increased by 38% from 2010 to 2011 (Associación Nacional de Empresas de

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[1] Rodriguez, Santana and Pardo (2015) include the data presented in this paper and a survey with motorcycle users in five Latin American cities.

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Fig. 1. Cities where data was gathered.

Table 1
Trips by mode for metropolitan regions.
Data Source: Urban Mobility Observatory (Corporación Andina de Fomento, 2010), except São Paulo (Metr,ô São Paulo, 2008).

Trips by mode	Bogotá	São Paulo	Recife	Barranquilla	Caracas	Buenos Aires
Motorcycle (%)	0.8	2.1	1.8	NA	0.6	4.1
Auto (%)	15.8	28.5	18.9	NA	23.8	42.1
Public transport (%)	60.5	29.0	35.0	NA	53.7	40.5
Taxi (%)	3.8	0.3	1.0	NA	2.5	4.3
Walk (%)	16.2	37.2	41.8	NA	18.2	8.5
Bicycle (%)	3.0	0.9	1.5	NA	1.3	0.5

Colombia, 2011).

Reasons for growth in motorcycle ownership and use include ease of acquisition and use, efficiency/economy, and enjoyment (Rogers, 2008). In Paris, people using motorcycles and scooters benefited from higher travel speeds: 46% and 127% faster than car and bus, respectively (Kopp, 2011). According to Vasconcellos (2008), the advantages of motorcycles in Brazil come from their ability to travel between lanes of automobiles, the ease of parking, and the low operational costs. Chang and Wu (2008) showed that younger motorcyclists with lower incomes registered the highest levels of dependence on that mode in Taipei. They suggest that a significant portion of users would choose motorcycles over public transportation, despite recent investment in Taipei's mass transit system. Chen and Lai's (2011) study of two major cities in Taiwan (Taipei and Kaohsiung) show that travel time is more important than travel cost when choosing the motorcycle mode. In Kuala Lumpur, Yamamoto (2009) found income to be negatively associated with motorcycle ownership and public transit accessibility, while distance from city center had a positive association. The lower cost of the motorcycles was also associated with increased demand for this vehicle in the UK (Duffy and Robinson, 2004). Vasconcellos (2005, 2013, 2008) suggests that supply and quality problems of public transport lead to increases in automobile and motorcycle use in Brazilian cities, and that motorcycle taxis function primarily where public transportation services are absent.

Much of the research into the use of motorcycles as delivery and taxi vehicles examines African cities. Kumar (2011) suggests that the collapse of state-run or contracted bus services in the 1990s led to the growth in motorcycle taxis in the in the capital cities Douala (Cameroon), Lagos (Nigeria), and Kampala (Uganda). Motorcycle mode share on selected road links was as high as 59% in Lagos, and 42% in Kampala. Guézéré (2015) shows how motorbike taxis came to dominate three secondary towns in Togo since appearing in the early 1990s After surveying motorcycle taxi growth in West and Central African cities, Diaz Olvera et al. (2012) conclude that public authorities must increase regulation of all transport modes. In Akuri, Nigeria, motorcycle taxis are very popular among users and highly profitable for operators (Fasakin, 2001). Motorcycle taxis are also popular in many Asian cities, including Jakarta and Bangkok, functioning despite their illegality in the latter city (Cervero and Golub, 2007).

Because of reduced travel times in urban areas versus public transport, and lower costs and greater mobility versus the automobile, income-generating activities of motorcycle taxi and urban delivery have grown in recent decades in Brazil (Silva et al., 2011). While Silva et al. maintain that growth in these activities have increased income and quality of life in Brazilian cities, Vasconcellos states that populist politics in Brazil emphasize job creation related to motorcycles while ignoring the social costs of this mode (2008). He also says urban delivery services on motorcycle have flourished in cities that have high rates of traffic congestion. Vasconcellos further notes that because well-paid work opportunities for young people with low education levels are scarce, motorcycle delivery work has become increasingly attractive. Sánchez Jabba (2011) examines motorcycle taxi activity in the Colombian city of Sincelejo, where motorcycle taxis are increasingly prevalent despite their illegality. His study concludes that although motorcycle taxi drivers engage in this activity because of higher earnings versus similar jobs, they would prefer jobs in the formal sector that have lower safety and health risks.

Deaths of motorcycle users represent a growing proportion of all road user deaths globally (World Health Organization, 2013), and motorcyclists are overrepresented in totals of traffic injuries and fatalities compared to other modes of travel. For example, while motorcycles represented 12.9% of the total fleet of motorized vehicles in São Paulo in 2011, motorcyclists accounted for 37.5% of all traffic fatalities, or 512 deaths of a total of 1365 (Companhia de Engenaria de Trânsito CET, n.d.). Further, while traffic deaths for all road users decreased in São Paulo by 23% from 2005 to 2013, with reductions ranging from 31% to 62% per mode, traffic deaths from motorcycles increased by 17% during the same period (Borges De Paula, 2014). One reason for this overrepresentation is the increased likelihood of injury or death of motorcycle users versus other mode users (Li et al., 2009). Vasconcellos notes that motorcycle traffic casualties in Brazil are treated fatalistically, as 'destiny' or 'the inevitable price of progress' (2005, p. 139).

In addition to the above research, at least one literature review on acquisition and use of motorcycles has been completed (Estupiñan et al., 2012). While most of the studies reviewed here use quantitative methods, a number of studies on motorcycles employ qualitative approaches (Huth et al., 2012; Tunnicliff et al., 2011). To

Table 2
Metropolitan regions and motorcycles per inhabitant.
Data Source: Urban Mobility Observatory, except Barranquilla (Observatorio Metropolitano, n.d.).

	Bogotá	São Paulo	Recife	Barranquilla	Caracas	Buenos Aires
Area (km^2)	2735	5302	2768	488	777	3830
Population	7,823,957	18,783,649	3,658,318	1,835,000	3,104,076	13,267,181
Motorcycles	116,433	652,225	78,192	NA	114,369	470,000
Motorcycles/1000 Inhabitants	15	35	21	NA	37	35

the best of our knowledge, however, the present project represents the only qualitative research on motorcycle use in Latin American cities. This article aims to contribute to transportation scholarship on Latin America by filling the current gap.

3. Method

To obtain a better understanding of the motivations that lead to the uptake of the motorcycle for urban trips, the ongoing employment of this mode and issues related to its use, the researchers conducted qualitative research. Qualitative methods can be particularly effective in illuminating travel behavior (Clifton and Handy, 2001). For example, Beirão and Sarsfield-Cabral (2007) used in-depth interviews to understand travelers' perceptions of private auto and public transportation, and data from focus groups helped researchers understand effective ways to promote active transport modes to young adults (Simons et al., 2014).

Qualitative methods allow researchers to gather rich data and conduct in-depth analysis, without using large volumes of numerical data or aiming to reach statistical significance (Bonilla-Castro and Sehk, 2005; Hernández Sampieri et al., 2010; Kerlinger et al., 2002). Following Van Manen (1990), Moustakas (1994) and Cresswell (2012), the researchers conducted a phenomenological study; that is, they used the narratives of individuals describing their lived experiences to discover the common meaning of a phenomenon (in this case, of using motorcycles for urban transport). According to Van Manen, when the phenomenological approach is employed, "research is a caring act: we want to know that which is most essential to being" (p. 5).

3.1. Sample

3.1.1. Cities

The six cities where researchers gathered data for this study were chosen for their diversity in nationality, geographical locations, and transportation system characteristics. The cities' transportation systems are characterized by relatively high mode share of public transportation (ranging from 60.5% of trips in Bogotá to 29% in São Paulo), and relatively small proportion of trips to work by motorcycle (ranging from 4.1% in Buenos Aires to 0.6% in Bogotá, see Table 1). The rates of motorcycles per 1000 inhabitants are similarly low – from 15 in Bogotá to 37 in Caracas (See Table 2).

The size of motorcycle fleets and the proportions of trips made by motorcycle for trips to work in these cities may be low compared to other modes; however, they have an important impact on the street-level experience. Motorcyclists are highly visible in all of the six cities, often agglomerating on road segments, intersections, or parking places. Just like other motorized vehicles, their engines and horns contribute to the urban cacophony.

Although there is no common indicator to rank these cities by their transportation problems, of the six, São Paulo and Caracas may have the worst problems with congestion. The former city has consistently broken its own records for traffic jams in recent years, with 344 km of congestion in 2014 (Reolom and Italiani, 2014), and researchers have noted the extreme congestion in the latter city (Lizarraga, 2012; Thomson and Bull, 2002). All six cities have some rail-based and/or BRT public transportation systems, but these can safely be characterized as insufficient in coverage relative to the demand for high-quality public transportation.

Motorcycles are relatively affordable for residents of Latin American cities. In Brazil, a motorcycle that is typically used by the participants in this study costs about 7400 reals, and purchasers can make monthly payments of 148 reals. With an average monthly salary of 2257 reals in São Paulo in September 2015 (Instituto Brasileiro de Geografia e Estatística, n.d.), this represents 6.6% of the monthly income. In Recife, where average incomes are considerably lower (1607 reals), this represents 9.2% of monthly income.[2] Monthly payments for new motorcycles are 29% of a minimum monthly wage (which a laborer or service worker can expect to earn) in Colombia and Argentina, and 19% in Brazil.[3]

The six cities have myriad regulations regarding the use of motorcycles (Rodriguez et al., 2015). License requirements vary, with more restrictions for younger, and/or motorcycles with larger engines. While Caracas has specific requirements for motorcycle taxis, these vehicles are prohibited in Sao Paulo, Bogotá and Barranquilla (although they operate informally in the latter city). Protective equipment, including helmets, is mandatory in all cities.

3.1.2. Study participants

The authors gathered data from a total of 121 participants, with about 20 participants in each city. Most participants in this sample used motorcycles for urban transportation, which have relatively small engines (between 125 and 200 cubic centimeters). However, eight participants in the original sample used larger motorcycles, primarily for recreational purposes (e.g., for motorcycle touring). These responses were excluded in this study because it focuses on motorcycle use for utilitarian urban trips. This left a total of 113 research subjects (see Table 3).

Study participants were recruited through snowball and convenience sampling (Creswell, 2012), both of which follow a non-probability sampling strategy (Bryman, 2004; Davidson, 2006). Although non-probability samples do not lead to representative or generalizable findings, they have yielded valuable results in diverse substantive areas, including transportation (Beirão and Sarsfield Cabral, 2007; Simons et al., 2014) and housing (Freeman, 2011). The authors feel that this sampling strategy was justified for the present project for several reasons. First, because there is little data available on the distribution of motorcycle users by type (e.g., delivery vs. private users) in Latin American cities, it would be difficult to ascertain the correct proportions of these users in a sample. Related to this, time and resource constraints made a non-probability sample the most appropriate method. Further, despite

[2] Of the six cities, recent estimates of average monthly salaries are readily available only for these two.

[3] The researchers found monthly minimum wages at government websites from Colombia, Brazil, and Argentina (Consejo Nacional et al., 2015; Ministerio del Trabajo, n.d.; Presidência da República, 2014, respectively), and prices of monthly payments for motorcycles on websites of motorcycle dealerships in each country (AKT Motos, n.d.; Consórcio Motos, 2015; Uno Motos, n.d., respectively).

Table 3
Participants by city and category of motorcycle user.

	Bogotá	São Paulo	Recife	Barranquilla	Caracas	Buenos Aires
Total #	20	18	21	22	20	20
Average age	33.5	29.4	29.1	37.4	30.4	38.4
Men (%) % Muj	90	89	62	100	50	90
Women (%)	10	11	38	0	50	10
Motorcycle taxi (%)	0	0	20	27	11	0
Motorcycle taxi user (%)	0	0	15	27	47	0
Delivery (%)	24	33	20	23	16	47
Private transport (%)	53	33	15	14	16	42
Selling motorcycle (%)	0	17	15	9	0	0
Potential user (%)	24	17	15	0	11	11

Table 4
Motorcycle owners and users engaged in commercial activity by participant category.

	Motorcycle taxi driver	Motorcycle taxi user	Delivery	Private transport	Selling motorcycle	Potential user
Motorcycle Owners						
Commercial activity						

the non-random sample, the exploratory nature of this project allowed the authors to identify topics that can be studied in a more systematic way in future research.

Together with the project sponsor (CAF), the researchers identified six categories of motorcycle users before commencing the data collection phase, drawing from previous research on motorcycle use and observation of transportation trends in Latin American cities:

1. Motorcycle taxi driver (these were not interviewed in Bogotá, São Paulo or Buenos Aires as they do not exist there, to the best of our knowledge)
2. Motorcycle taxi passenger (same as above)
3. Motorcyclist for goods delivery
4. Motorcyclist for private transportation, for individual use or for multiple users (e.g., with family members or friends)
5. Owner in the process of selling motorcycle
6. Potential motorcyclists (people that intend to acquire motorcycles)

Because there are little data available on the distribution of motorcycle users by type (e.g., delivery vs. private users) in Latin American cities, and time and resource constraints, the authors did not attempt to attain a representative sample of motorcycle users. The participants' average age was 32.9 years old (Standard Deviation 9.6) and 19% were women and 81% men. Women participants were most strongly represented in the categories of “motorcycle taxi user,” “motorcyclist for private transportation” and “potential motorcyclists.” The table below provides descriptive statistics for respondents.

There was overlap between the categories and people that owned motorcycles and motorcycle users concerned by a commercial activity. The former were represented in the categories “delivery,” “motorcycle taxi driver,” “private transport,” and the latter in “motorcycle taxi driver,” “motorcycle taxi user,” and “delivery” (see Table 4).

The majority (57%) had not completed high school or had obtained no more than a high school education. Most participants (70%) were employed in the service sector, including delivery workers, doormen, hair stylists, security guards, and retail workers. Although the researchers did not ask for income specifically, it is safe to assume that participants with low levels of education and employed in service occupations belong to low-income populations. The participants in professional occupations (30%) included publicists, civil servants, teachers, a doctor and an engineer. Most of the professionals were interviewed for the categories “motorcyclist for private transportation” and “potential motorcyclists.”

3.2. Procedure

To ensure uniformity in data collection, the authors created a protocol for interviews and focus groups in the six cities. They created the protocol together with the project sponsor (CAF) and before beginning data collection. Data was gathered over an eight-month period (February to October 2012). Interviews were conducted in Spanish, and in Portuguese in Brazilian cities, lasted between approximately 40 and 120 min, and were concluded when no new themes emerged in most cases, or when interview subjects needed to leave in a minority of cases.

The researchers used various methods to recruit interview and focus group participants, including targeting businesses that use motorcycles for delivery, contacting a motorcycle advocacy organization,[4] using researcher networks and electronic social media, and approaching potential interview subjects at places where motorcyclists congregate (e.g., motorcycle taxi stands). The researchers constituted focus groups when they encountered groups of motorcycle users, at which time the researchers would ask the motorcycle users to participate in the study. The study includes a total of 12 focus groups, ranging from 3 to 6 participants, with an average of 4 participants in each focus group. The researchers gathered data in individual interviews from 58% of the participants and in focus groups from 42%. These techniques were used in different proportions in each of the cities (see Table 5).

In São Paulo, Recife, Barranquilla and Caracas, researchers contracted one local research assistant (except for São Paulo, and

[4] São Paulo has a “Union for Motorcycle Delivery Workers,” or “Sindicato dos Motoboys” in Portuguese.

Table 5
Proportion of participants in interviews or focus groups by city.

	Bogotá	São Paulo	Recife	Barranquilla	Caracas	Buenos Aires
Interview (%)	53%	83%	60%	9%	100%	56%
Focus group (%)	47%	17%	40%	91%	0%	44%

Barranquilla, where two were contracted) to recruit interviewees, setting up interviews, conduct interviews and take notes, and transcribe notes. A total of seven research assistants helped gather the data, and in the case of two cities (Caracas and Buenos Aires), gathered all data used for this paper.

The researchers recorded the interviews and focus groups with electronic devices in addition to taking notes. Interviews and focus groups were conducted in very varied premises and often in informal settings, including restaurants and public spaces. The interview guidelines included ten open-ended questions that were asked of all user categories (except those who had never owned a motorcycle). These questions included “When did you buy a motorcycle instead of another vehicle? ” and “how satisfied are you with the motorcycle as a form of transportation? ” Researchers asked additional open-ended questions tailored for each specific category of motorcycle transport user; for example: “Why do you use motorcycle taxis rather than other forms of transportation? ” The researchers gave participants freedom to provide information they thought was most relevant, and to do so in their own words. For their participation, interviewers gave respondents compensation equivalent to ten US dollars.[5]

The researchers transcribed the notes from the interviews and focus groups and consulted the recordings to clarify participant responses when necessary. In the analysis phase, the authors went through the data to highlight significant statements, placed these statements into common themes among cities and participants, and then clustered these themes into domains. The themes were judged by the authors to most accurately reflect the data gathered and the importance of the topics to the participants.

4. Results

The researchers identified themes emerging from the data that were most commonly mentioned in all six cities, and organized these themes into four domains. The first domain encompasses transportation-related reasons that participants began to use motorcycles and continue to choose this travel mode for their daily transport needs. The second is related specifically to economic motivations for using motorcycles. The third domain contains themes related to the risks associated with motorcycle use. Finally, the authors present cultural issues that emerged in the data.

4.1. Transportation-related motivations for motorcycle use

4.1.1. Speed/time savings

The main reason participants gave for first buying motorcycles and continuing to travel by this mode was the ability to travel at high speed and save time in transport. Some participants reported using the time saved in travel for other activities generally, others for leisure activities specifically.

> “*Time is money and quality of life. I sleep more because I don't have to wake up early to go to work; I save one and a half hours on my way to work, and one and a half hours on my way home. I use that time to take my time when I shower, read the paper, be with my family and sleep*.”

-Motorcycle delivery worker, São Paulo, male, 31.

Participants spoke of the small amount of space the motorcycle requires. Some specifically mentioned bypassing congested traffic by travelling between lanes of slow-moving autos (these spaces were called “corridors” by some participants) or on the shoulder. Participants also frequently mentioned the ability to park with relative ease. Both provided time savings for motorcycle users.

> *The motorcycle lets you go wherever you want quickly, riding between cars stuck in traffic, park in practically any place, no matter how small. You save a lot of time.*

-Potential user, Recife, male, 26.

4.1.2. Modal choices - public transport

In general, participants had negative views of public transport. Words participants used to describe public transportation include: “chaotic,” “dirty,” “dangerous” [from crime], “unreliable,” “slow,” and “impossible.” Some said they would not use this mode, and many vehemently expressed their dislike:

> *I used public transport once and I hated it.*

-Motorcycle delivery worker, São Paulo, male, 27.

Participants mentioned crowded vehicles:

> *The problem with public transportation is that people travel like cattle.*

-Private user, Buenos Aires, male, 21.

Some mentioned that service hours did not suit their schedules:

> *Public transportation is bad and I get out of work late, at 10 pm, so I would have to wait a long time for a crowded bus.*

-Private user, Recife, female, 28.

Some mentioned “flexibility,” the ability to bundle trips and travel without transferring, as an advantage of the motorcycle over public transportation.

Motorcycle taxi users said this mode was faster than public transport, and worth the investment because of time savings. In Barranquilla, passengers mentioned making intermodal trips (motorcycle taxi + bus), and said they would probably use the BRT system (Transmetro, implemented in 2010) if the routes were closer to their homes.

Only one participant mentioned using public transportation instead of a motorcycle on rainy days.

4.1.3. Modal choices - automobile

Participants expressed diverse views on auto ownership. For some, owning a car was an aspirational goal, something very desirable but unattainable given their current income levels. For example, motorcycle taxi passengers in Barranquilla ranked private cars as their first transport option, if money were not an issue. Of participants that aspired to own cars, many said that although they would use these vehicles, they would still keep their motorcycles for some trips, or for most travel. Other participants showed no desire to own autos.

Many participants that aspired to own cars viewed them as vehicles that “are for the whole family,” versus trips that can be made alone. Some, including delivery workers, reported owning

[5] In the case of São Paulo and Recife, this value was higher (40 Brazilian Reals or about $21.70 in April 2012), to reflect higher costs of living in Brazil. Examples of this compensation include cellphone credits, book store gift cards, and a paid meal at the restaurant where the interview took place. Researchers consulted participants as to their preferred form of compensation.

and using cars, but used these mainly with their families on their days off.

Respondents that owned a car and a motorcycle often expressed a very strong preference for motorcycles over cars:

> *When I am leaving home with my car, I think ten times before I grab the keys.*

-Delivery worker, São Paulo, male, 31.

Many participants (particularly from Caracas, São Paulo and Bogotá, and to some degree, Buenos Aires) said that congestion severely limited the effectiveness of trip-making by auto:

> *Cars are useless in this city.*

-Potential motorcyclist, Bogotá, male, 38.

Compared with motorcycles, autos were universally seen as safer in terms of exposure to traffic injury:

> *A disadvantage of the car is that on a motorcycle you can't lose concentration for a single minute, but in a car you can.*

-Private user, Buenos Aires, male, 40.

4.1.4. Modal choices - bicycle and walking

Participants often referred to bicycles as an inexpensive and environmentally friendly mode. Some saw bicycles as an advantageous mode for shorter trips or had used them in intermodal trips with public transportation. Walking was mentioned only briefly in Caracas and Barranquilla.

4.1.5. Road infrastructure

Some participants mentioned that badly maintained roads posed a danger in their cities. Others said that additional space for designated motorcycle parking was needed. Exclusive lanes for motorcycle travel were seen as possible solution by some participants. Where exclusive lanes do exist, participants cited problems, e.g., congestion (São Paulo) and blockage by trash and public transportation (Barranquilla). In Barranquilla, participants said that the exclusive lanes "never work," and that "nobody uses them," although they also said that some cautious riders did use these lanes. Participants from Buenos Aires admitted that they frequently used dedicated bicycle and public transportation lanes.

4.2. Economic motivations

4.2.1. Cost

Participants in most cities stated that the low cost of buying, using and maintaining motorcycles was a fundamental factor for adoption and continued use of the mode. The ability to access credit and make small monthly payments was mentioned in some cities (Bogotá and São Paulo), as were low fuel and maintenance costs, and affordability compared to other modes, especially automobiles and public transportation.

> *With public transportation I spend R$300 [US$ 164]*[6] a month [to travel to work]. With a motorcycle I would spend R$200 [US$ 109].

-Private user, São Paulo, male, 34.

> *Motorcycles are much cheaper than public transportation or cars, and there are many ways of financing that make buying one easy that require only two "SMMMLV" ([minimum monthly salaries]). And monthly payments and maintenance costs are low.*

-Potential user, Bogotá, male, 40.

Lower cost of parking was also mentioned as an advantage versus cars:

> *I pay for parking three times a day: at the university, at work and at home. With a motorcycle, I wouldn't pay anything.*

-Potential user, São Paulo, female, 25.

However, participants in Buenos Aires mentioned the rising cost of maintaining motorcycles. Motorcycle taxi users said this mode was more expensive than public transport (and cheaper than conventional taxis), but the mobility advantages were worth the extra cost:

> *Even though it can be more expensive than the bus, motorcycle taxis are faster and bring you exactly where you want to go. By bus, you have to walk to the stop, and to wherever you are going.*

-Motorcycle taxi user, Recife, female, 27.

4.2.2. Employment opportunities

Employment opportunities were most important to those that made a living by using this transport mode — motorcycle taxi drivers and motorcycle goods delivery workers. Owning, or at least having access to, a motorcycle was a requirement to work as a motorcycle taxi driver or delivery worker. The researchers termed this the "bicycle thief effect," alluding to the 1948 film by Vittorio de Sica (1972). The workers indicated that owning a motorcycle was a way (some specifically said the best way) to increase their income. Participants in São Paulo, Bogotá and Barranquilla mentioned that for low-income workers, having a motorcycle could be as effective as education or work experience for increasing income.

> For someone from the periphery [poor neighborhood far from city center], this is the best job you can get.

-Delivery worker, São Paulo, male, 25.

Some motorcycle taxi drivers said they enjoyed having flexible work hours.

4.3. Risk

4.3.1. Vulnerability to traffic injury/death

Participants almost universally emphasized their vulnerability. The idea that the motorcycle driver or passenger him/herself was the "chassis" of the vehicle was specifically stated in three different cities and two countries:

> *Your body is the chassis.*

- Delivery workers, São Paulo, male, 27 and 29, and private user, Recife, female, 26.

> *We [motorcycle taxi passengers] are the chassis.*

-Motorcycle taxi passenger, Barranquilla, male, 52.

"Bumper" was another metaphor for vulnerability:

> *My chest is the bumper.*

-Potential user, Bogotá, male, 38.

> *You are the bumper.*

-Delivery worker, Buenos Aires, male, 30.

Participants in almost all cities perceived that motorcycles put users in an inherently vulnerable position. In Barranquilla a participant could not imagine how motorcyclists' vulnerability to traffic injury/death could be reduced. A participant in Recife said the motorcycle was a "weapon."

Many participants reported seeing fatal or severe collisions, having one or several injuries themselves, or having family

[6] This US dollar equivalent is from April 2012, when the interview was conducted.

members or friends that died or had sustained severe injuries in traffic. However, most of these did not say that this motivated them to stop using this mode. Some spoke in fatalistic terms about traffic casualties:

It's like smoking, you know it will kill you, but you keep on doing it. You think it won't happen to you; 'look at that guy, he was careful his whole life and died of cancer.'

-Private transport user, Buenos Aires, male, 48.

However, some participants displayed a more cynical attitude:

On a motorcycle, you get places faster, and you die faster too.

-Motorcycle delivery worker, São Paulo, male, 29.

The above quote also emphasizes the tradeoff between mobility and safety, which was very clear for many participants, including motorcycle taxi users:

I have been involved in some accidents, but in general, it's worth it.

-Motorcycle taxi user, Recife, female, 27.

Some had a seeming disconnect about the possibility of traffic injury/death and their own vulnerability; e.g., one aspiring motorcyclist had a cousin that died in a collision but nonetheless perceived the motorcycle as a good option for urban travel. Other participants mentioned that vulnerability to traffic injury/death increases when it is raining, due to slippery road conditions.

4.3.2. Dangerous/safe driving

Participants mentioned both dangerous and safe driving practices in the context of the behavior of motorcyclists and other road users. Some said that driving between lanes of congested auto traffic was dangerous. Newer motorcyclists were perceived to drive in more dangerous ways:

This city is full of jerks that believe they already know how to ride.

-Delivery worker, Barranquilla, male, 33.

A few participants regretted having driven dangerously in the past, and many mentioned the benefits of driving in a prudent manner, at lower speeds, and of paying close attention. Users in São Paulo had a specific term for prudent driving: "drive 'in a good way'" (*andar 'de boa'*). Some participants mentioned groups that drive more carefully: women, older people, and motorcyclists that were transporting their wives and/or children.

Participants specifically mentioned that other motorists of other vehicles drove in dangerous ways, including bus and (automobile) taxi drivers in Barranquilla, cars and especially trucks in São Paulo, buses in Caracas, and all road users in Buenos Aires.

4.3.3. Use of protective elements

Most participants reported using safety equipment (clothing and equipment mounted on vehicles), and displayed varying degrees of enthusiasm for these. Some also indicated that they used the equipment with greater conviction after experiencing a crash. Participants from Recife, São Paulo, Bogotá and Caracas emphasized the importance of using safety equipment. In Buenos Aires, some said the equipment was a requirement that was imposed on them:

It takes me longer to put on my helmet than ride the 15 blocks to make a delivery.

-Delivery worker, Buenos Aires, male, 21.

While another said safety equipment was essential:

Helmets save your life.

-Private user, Buenos Aires, male, 48.

Improper use of safety equipment was also highlighted:

They [motorcycle riders] wear helmets but they don't fasten the strap.

-Potential user, Bogotá, female, 42.

Interestingly, many participants from São Paulo and Recife mentioned that their motorcycles were fitted with a "cut-kite" ("corta-pipa"), an antenna-like rod that cuts glass-covered kite strings that often hang in roadways, known to be deadly to motorcyclists.[7]

4.3.4. Other health issues

Participants from Bogotá, São Paulo and Buenos Aires mentioned problems related to air quality and soot:

My eyes get red from the pollution. My face is totally dirty and when I take a shower, the water is black.

-Delivery worker, São Paulo, male, 27.

You have to take two showers [to get rid of soot].

-Delivery worker, Buenos Aires, male, 34.

Back pain and problems with kidneys were mentioned in Barranquilla, and in several cities participants mentioned that the "stress" of riding motorcycles in traffic was a health problem.

4.3.5. Crime

The most prominent topic regarding crime was motorcycle theft. For motorcycle taxi and delivery workers, who rely on these vehicles to make a living, this issue was very serious, and many workers mentioned their motorcycles having been previously stolen. In some cities, participants also spoke of robberies perpetrated by motorcycle-riding criminals. Such groups were said to ride in pairs, with a driver and a passenger that bears a weapon to hold up robbery victims. In São Paulo, a delivery worker mentioned that he delivered pizza for a period, but stopped because of the risk:

Motoboys [motorcycle delivery workers] get mugged a lot. For example: assailants will order a pizza delivery and steal the motorcycle, and if you react, you can get killed.

-Delivery worker, São Paulo, male, 29.

Some participants stated that motorcycle users were as exposed to crime as any other mode user; others perceived higher exposure for this mode.

4.3.6. Discontinuing motorcycle use

Some participants intended to sell their motorcycles and stop using the mode altogether. They predominantly cited injuries sustained or near-death experiences while riding as reasons; for example:

I knew for sure that I would stop riding when I was on Avenida Tiradentes and a car swerved to avoid a pothole and hit me, and I hit a [public transport] mini-bus. I was lucky that it stopped; if it didn't I believe I could have died.

-Selling motorcycle, São Paulo, male, 28.

One woman was pregnant and acquiesced to her family's pleas for her to discontinue motorcycle riding because of the risk of traffic injury or death. Other users stated said they would discontinue use if they had children. Some perceived that safety had worsened, motivating them to stop using the mode.

[7] Kite-fighting is a popular past-time in Brazil, and opponents attempt to cut each other's' kite strings, which are covered in sharp glass shards. The strings of cut kites often hang across roadways, posing a deadly danger to passing motorcyclists.

4.3.7. Satisfaction/recommend use to others

Many participants were very satisfied with motorcycles and enthusiastically recommended using these vehicles to others. Some indicated gender or age as determining factors, specifically mentioning that they would not want their children and/or wives using this mode. Other participants emphatically stated that they would not recommend motorcycle use, primarily because of the vulnerability to traffic injury/death.

4.4. Cultural aspects

4.4.1. Feeling of liberty

Participants in all cities mentioned the feeling of freedom that they associated with the motorcycle mode. For some, this feeling was one of several advantages associated with the mode, while others saw this as its fundamental. Participants spoke of "adrenaline," "wind in your face," "independence," and relaxing sensations.

4.4.2. Identity of motorcyclists

For participants in all cities, using the motorcycle for urban travel was more than a simple mode choice; using this mode defined who they were in a larger sense. Many mentioned a sense of solidarity among motorcyclists, and a mutual care for each other's well-being vis-à-vis users of other modes, e.g., cars and trucks:

We are all brothers… it's very strange. When there is a motorcycle accident, all the riders stop and stay there. Hit and run is very common.

Delivery worker, Buenos Aires, male, 30.

"*We take care of each other, we look out for people in our 'union,'*"

Motorcycle taxi driver, Barranquilla, male, 36.

Motorcycle taxi and delivery workers mentioned forming friendships among colleagues and using motorcycles for recreation outside of motorcycle-related work. In São Paulo, this identity was related to the place that they lived, the "periphery" (low income neighborhoods, often informal, outside of the central city), where they said it was common to see motorcycles parked outside of houses.

Some respondents also identified with the vehicle itself, reporting that it had become an extension of their bodies:

You become one with the motorcycle.

-Private user, Barranquilla, male, 25.
(Similar statements from users in all cities).

Others expressed feelings of adoration for their vehicles:

I call her 'saint.'

-Delivery worker, São Paulo, male, 29.

4.4.3. Stigmatization of motorcyclists

Many participants, and particularly delivery workers, mentioned that they felt stigmatized and were discriminated against by other road users:

The biggest danger is the lack of caution of car and truck drivers, who don't respect motoboys [motorcycle delivery workers].

-Delivery worker, São Paulo, male, 31.

In Caracas, participants perceived robberies by motorcycle-riding perpetrators as the cause of this stigma and felt that they suffered unjustly, as they did not commit such crimes themselves:

Motorcycles are fast so they are used to rob people, and 'good people pay for sinners [pagan justos por pecadores]'.

-Potential user, Caracas, female, 42.

4.4.4. Culture of road use

Participants in all cities mentioned the need to establish a better road culture to reduce collisions, and often used words like "anarchy" to describe existing conditions. Many saw a lack of enforcement of existing traffic laws and education as a cause of a "free for all" culture on roads.

4.4.5. Social pressures

Family members–particularly mothers–were cited as being against this mode of travel because of the risks related to traffic casualties:

My mom prayed every day that I would stop riding.

-Selling Motorcycle, São Paulo, male, 23.

My family told me 'I would rather give you a coffin than a motorcycle.'

-Private user, Barranquilla, male, 25.

Many participants mentioned uptake of this mode in the context of a shared activity with neighborhood friends, all of whom began to use motorcycles together. One participant said that his entire family used these vehicles for transport and encouraged him to adopt the mode.

4.4.6. Gender issues

Although the authors do not claim to have reached a representative sample, it is worth noting that 57% of motorcycle taxi users were female, while most of the participants the researchers interviewed were male (81%). One woman in São Paulo felt that she was "breaking a taboo" by being a woman on a motorcycle. Many participants associated safer riding with women.

4.4.7. Importance of education/lack of formal education

Most informants had learned to ride informally, often with friends, and many said that more formal and rigorous education was needed to make this mode safer. Participants that reported safe driving had learned this behavior simply from experience, without formal instruction. Participants from Bogotá said that the requirements for getting a license should be more rigorous, and some from São Paulo said that while one previously didn't need a license to use this mode, it was now necessary. In Barranquilla, many participants said they obtained the licenses in informal or illegal ways.

4.4.8. Seduction

Some participants cited the motorcycle as a way to attract members of the opposite sex, particularly men vying for women's attention:

The motorcycle makes your stock go up… if they [women] see you walking they might not give you the time of day.

-Private user, Buenos Aires, male, 30.

Another participant from Buenos Aires mentioned that women displayed greater interest when he was riding without the box that he uses to transport goods for delivery. One woman motorcyclist mentioned flirting from men in other vehicles and insinuated that she enjoyed this attention:

My motorcycle is pink. Guys honk at me, they flirt. (laughs)

-Private user, São Paulo, female, 24.

Table 6
Perceived advantages and disadvantages of travel modes.

	Motorcycle	Auto	Public Transportation	Bicycle	Walk
Mentioned in Advantages	All cities – High travel speed / bypass congestion – Easy to park – Carry people and objects – Low cost (purchase and operation) – Source of income – Flexible work schedule – Feeling of freedom – Identity/belonging to a social group – Seduction	All cities – Safe – Can bring family – Protection from weather	All cities – Safe – Protection from weather – Can sleep on	All cities – Inexpensive – Good for environment – Healthy – Fast for short trips – Can combine with public transportation	Caracas, Barranquilla – Good for family outings
Disadvantages	– High risk of traffic injury/death – Vehicle can get stolen – Exposure to robbery – Exposure to air pollution – Other health issues (back pain, kidneys, stress) – Exposure to inclement weather – Stigmatization by other mode users – Social pressures to stop using	– Slower (due to congestion) – Hard to find parking – Expensive	– Slow – Expensive – Uncomfortable – Crowded – Dirty – Chaotic – Unreliable – Insufficient network coverage – Insufficient operating hours – Long waiting times – Have to make transfers	– Not suited for long trips – Physically demanding – Not safe (from traffic)	– Not mentioned

5. Discussion

The advantages and disadvantages participants ascribed to motorcycle, auto, public transportation, bicycle, and walk modes are listed in Table 6. The motorcycle has many valued attributes, including mobility factors such as fast travel speeds and low costs – items that are highly valued in a typical 'derived demand' model of transportation (Small and Verhoef, 2007). However, participants also mentioned attributes that are related to the visceral experience of riding these vehicles, such as feelings of freedom and exhilaration, and becoming one with the motorcycle. Socially embedded (Granovetter, 1985) advantages, such as those related to identity and belonging to a social group, as well as seduction, were also important. Participants also mentioned serious disadvantages of motorcycles, including those related to public health (exposure to traffic injury/death, air pollution, back and kidney pain, stress), crime, inclement weather, and social issues, such as stigmatization.

Non-motorcycle modes offered some advantages, but their disadvantages were generally more compelling to participants. Both auto and public transportation are associated with high costs and reduced mobility (lower speed and higher cost). Some participants showed some interest regarding the possibility of bicycles for urban travel, at least for shorter trips.

Although the bulk of this analysis focuses on topics that were common to all six cities, it is worth mentioning some of the differences by city. Although congestion was a primary motivating factor for motorcycle use in all cities, participants most vehemently emphasized this factor in Sao Paulo and Caracas, both of which may have the most severe congestion of the six cities. Relative to the other cities, participants in Barranquilla saw safety equipment as relatively unimportant, and emphasized the discomfort that it causes. Perhaps not coincidentally, Barranquilla has the consistently hottest climate of the six cities. Further, while participants in other cities mentioned that acquiring licenses to use motorcycles via official means was becoming more commonplace, participants in Barranquilla reported acquiring these in informal ways.

Some aspects of motorcycle use varied by type of user. Motorcycle delivery or taxi workers belonged to lower-income groups, and identified themselves as different from other types of motorcycle users (e.g., those that used motorcycles for transport). Respondents with lower levels of education and in lower-paid jobs emphasized stigmatization of motorcycle users, and said that other traffic users singled them out on the streets, exposing them to increased danger. These users also emphasized solidarity among, and identifying as, motorcycle users, which more affluent participants did not mention.

5.1. Validation of findings and emerging topics

While the results of this study support many of the findings of the literature on motorcycle use reviewed by the authors, the data of this study show some important differences, and also uncovered issues related to motorcycle use not encountered in the literature review. These emerging topics should be researched further and taken into account by policymakers who wish to address motorcycle use in Latin American cities.

Existing literature cited the reduced travel time of motorcycles as a primary motivation for motorcycle use (Chang and Wu, 2008; Kopp, 2011; Vasconcellos, 2008), a finding that was clearly validated by this research. Previous research found the lower cost associated with motorcycle purchase and operation to be fundamental to the choice of this mode (Duffy and Robinson, 2004; Silva et al., 2011; Yamamoto, 2009), however, the present study shows a less clear relationship between cost and motorcycle use. While some informants indicated that they would use an automobile if they could afford it, others said they would likely continue to use motorcycles even if they could afford a car. Some reported owning a car but using the motorcycle for most urban travel, and strongly preferring the latter mode. This is perhaps the most important emerging issue: many participants did not appear to perceive motorcycles as a step toward automobile ownership. This suggests that as incomes rise, motorcycle use will not necessarily decrease, even among people that can afford to buy cars. This result confirms Chen and Lai's (2011) finding that travel time is more important than travel cost when choosing the motorcycle as a travel mode.

Participants praised motorcycles as having important advantages that corresponded to perceived failures of public transportation (e.g., fast vs. slow, affordable vs. expensive, comfortable vs. uncomfortable, etc). This confirms literature that cites poor public transportation as an important reason for motorcycle mode growth (Kumar, 2011; Vasconcellos, 2005, 2013, 2008; Yamamoto, 2009). Also, literature suggested that auto congestion was an important motivation for motorcycle use (Vasconcellos, 2008), and this was confirmed by participants in this study, particularly in the cases of São Paulo, Bogotá and Caracas.

In particular, it appears that traffic congestion contributes to the flourishing of motorcycle delivery services, as Vasconcellos (2008) claims. Further, the economic opportunities provided by motorcycle-reliant jobs (Sánchez Jabba, 2011; Silva et al., 2011) were crucial to workers that participated in this study.

The vulnerability of motorcycle users to traffic death/injury (Haworth, 2012; Huth et al., 2014; Kopp, 2011; Vasconcellos, 2013, 2008; World Health Organization, 2013) was strongly reflected in statements by participants in this study. While the fatalistic attitude toward these deaths noted by Vasconcellos (2008) was demonstrated by many participants, some of them displayed outright cynicism regarding this vulnerability ("… you die faster…"). Participants' acute awareness of their vulnerability was also reflected by the statements that their body is the "chassis" and absorbs impacts from crashes.

While motorcyclists' increased exposure to traffic-related injury and death has been documented, participants mentioned other health issues not encountered in the literature, including eye irritation and dirtied faces from local pollutants (which may also be associated with respiratory illnesses), back and kidney pain, and general stress from riding in traffic. These emerging issues may have important detrimental health effects in the short, medium and long term.

Participants' statements also made clear the socially embedded nature of motorcycle adoption and use. Many of the social aspects of motorcycle use were positive, including the association with friendship, a sense of belonging, and seduction. However, participants also mentioned negative social aspects of using the mode, such as the stigmatization of motorcyclists by other road users, and family pressures to use safer travel options.

Other emerging issues participants mentioned include the lack of enforcement of traffic laws and dangerous driving, motorcycle user education and licensing practices, the adoption of prudent motorcycle driving practices, motorcyclists' exposure to crime, factors leading to the discontinuation of motorcycle use, and the employment advantages of motorcycle ownership versus further education.

5.2. Policy implications and conclusion

Because of the non-representative sample, the authors make no pretense of statistical power or generalizability of the results. Nevertheless, the participants' perceptions highlight issues that may be actionable for transportation planners and decision makers in other Latin American cities. The most pressing issue is the need to make motorcycle transport safer, as indicated by the participants' view that they are highly susceptible to death and injury from traffic collisions. This perception is corroborated by empirical evidence (Chandran et al., 2012; World Health Organization, 2013). Possible strategies to make traffic safer for motorcyclists include slower speeds on city streets, since collision severity increases as speeds increase (Anderson et al., 1997; Haddon Jr, 1980).

Study participants spoke of lax licensing and rider training, as well as inconsistent use of protective elements. Stricter licensing procedures and effective rider training may contribute to safety (Baldi et al., 2005; Lin and Kraus, 2009). Latin American cities may benefit from more comprehensive regulation and enforcement regarding motorcycles in general. In the present sample, Barranquilla appeared to be on the extreme end of the spectrum with respect to informality.

Safe driving habits, which participants learned informally, may be an important area for motorcyclist training. Additionally, policymakers could seek to curb dangerous driving habits for all road users and encourage a culture of street safety. This may be particularly important for lower-income motorcycle users, who perceived elevated exposure to dangerous driving. These users may also benefit from measures to increase use of safety equipment. Community participation, which has proven to be effective in increasing helmet use in Thailand (Ratanavaraha and Jomnonkwao, 2013), may be an effective way encourage more consistent use of protective elements.

Beyond making use of motorcycles less dangerous, transportation planners in Latin America could also explore ways to provide safer mobility options. While autos are inherently safer than motorcycles, they lead to congestion (Santos et. al, 2010; Thomson and Bull, 2001) and have a host of other undesirable effects for public health and the environment. As such, transportation planners could seek improvements to public transportation to meet the needs of present and potential motorcycle users, and explore possibilities for bicycle transport, which was mentioned as a viable option for shorter trips by participants in all cities. Stimulating both of these modes can help create healthier, more environmentally sustainable and socially equitable cities (Banister, 2008).

Participants overwhelmingly had negative views of public transportation in their cities; many researchers (Estache and Gómez-Lobo, 2005; Gwilliam, 2003; Hidalgo and Huizenga, 2013; Vasconcellos, 2005, 2014) corroborate this view of deficient public transportation provision in urban areas of Latin America. Public transportation must improve in these cities if it is to compete with the mobility and cost benefits of motorcycles. While public transportation's mobility benefits will not likely reach those of motorcycles (e.g., door-to-door service), travelers may prefer to use the former mode based on other trade-offs, such as safety, particularly if public transportation services are improved. Options to achieve this include increasing frequency, service coverage, operating hours, and cleanliness, as well as implementing exclusive rights of way on main transportation corridors. Once public transportation provision is improved, marketing efforts, including fare-free periods and publicity campaigns, can help attract new users (Beirão and Sarsfield Cabral, 2007; Redman et al., 2013).

Due to the high rate of injury and death related to motorcycles, the social cost of this mode is elevated. To internalize these externalities, the cost of using motorcycles can be increased. This would also make safer transportation options more competitive relative to the motorcycle. Equity concerns related to this approach could be addressed by lowering fares and improving levels of service of public transportation. Options to increase the cost of motorcycle use are the same as those for increasing auto costs. These include higher costs for purchasing vehicles and registration fees, higher fuel costs, road and parking pricing, higher insurance rates, and integrated land use and transportation planning (Banister, 2008; Buehler, 2011; Cameron et al., 2004; Shoup, 2005).

Implications for policy in this section emphasized measures that all six cities in this sample would benefit from. The main difference between the cities was the existence of motorcycle taxi services (not present in Sao Paulo, Bogota and Buenos Aires). Motorcycle taxis in Latin American cities require further research; although the mobility benefits for customers, and economic benefits for drivers, appear to be large, there may be considerable risks associated with this mode, especially in terms of traffic injury. Because of the potential for increased risk of traffic injury, the authors do not suggest that motorcycle taxi services be implemented where they do not exist today.

The findings of this paper suggest that because of their many advantages (particularly in terms of travel time) the mode share of motorcycles may continue to increase in the near future if current transportation scenarios persist in Latin American cities. Despite users' increased exposure to traffic injury/death, participants' statements suggest that the lack of competitive mobility options lead them to continue to use this mode, regardless of its increased danger. Significantly improving public transportation, and possibly bicycle transport, may be challenging politically and capital-intensive in the short-term. Improving driver education, licensing procedures, and enforcement for greater road safety may require lower capital investments and have shorter implementation times, and could quickly deliver important benefits to motorcycle users in terms of traffic safety.

Although the results of this study cannot be interpreted as representative in a statistical sense, the authors feel that the broad range of data gathered validated the exploratory, qualitative approach this research employed. The flexibility of the open-ended interview format allowed participants to include a greater range of topics and richer data than a more restrictive procedure (e.g., a survey) would have permitted. Participants brought up topics that the researchers had not previously considered in the questions, such as the feeling of freedom and cultural aspects of motorcycle use, which are important to understanding the factors that lead to the uptake and continued employment of this mode, and can be explored in future research.

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References

AKT Motos. (n.d.). Simulador de crédito. Website of motorcycle dealership, Bogotá, Colombia. Retrieved from 〈http://www.aktmotos.com/compra/simulador-de-credito〉.

Anderson, R.W.G., McLean, A.J., Farmer, M.J.B., Lee, B.-H., Brooks, C.G., 1997. Vehicle travel speeds and the incidence of fatal pedestrian crashes. Accid. Anal. Prev. 29 (5), 667–674.

Associación Nacional de Empresas de Colombia, 2011, March 2. La Industria de Motocicletas en 2011. Press Release. Retrieved from 〈http://www.andi.com.co/cinau/Aos%20Anteriores%20Moto/Comunicado%20de%20Prensa%202011.%20Mercado%20de%20Motocicletas.%20Enero%20-%20Diciembre.pdf〉.

Baldi, S., Baer, J.D., Cook, A.L., 2005. Identifying best practices states in motorcycle rider education and licensing. J. Saf. Res. 36 (1), 19–32, http://doi.org/10.1016/j.jsr.2004.11.001.

Banister, D., 2008. The sustainable mobility paradigm. Transp. Policy 15 (2), 73–80, http://doi.org/10.1016/j.tranpol.2007.10.005.

Beirão, G., Sarsfield Cabral, J.A., 2007. Understanding attitudes towards public transport and private car: a qualitative study. Transp. Policy 14 (6), 478–489, http://doi.org/10.1016/j.tranpol.2007.04.009.

Bonilla-Castro, E., Sehk, P.R., 2005. Más allá del dilema de los métodos: la investigación en ciencias sociasles. Editorial Norma. Retrieved from 〈https://books.google.com/books?hl=en&lr=&id=REOIWoQuAL4C&oi=fnd&pg=PA17&dq=bonilla-castro&ots=Ty5pqM2tH6&sig=3ZAfZE2QAVMLztTpfOnZOnObAPY〉.

Borges De Paula, M.E., 2014. Evolução do número de mortes no trânsito em São Paulo. Companhia de Engenaria de Trânsito Nota Técnica 232Nota Técnica 232, São Paulo. Retrieved from 〈www.cetsp.com.br/media/294272/nt_232.pdf〉.

Bryman, A., 2004. Social Research Methods, 2nd ed. Oxford University Press.

Buehler, R., 2011. Determinants of transport mode choice: a comparison of Germany and the USA. J. Transp. Geogr. 19 (4), 644–657, http://doi.org/10.1016/j.jtrangeo.2010.07.005.

Cameron, I., Lyons, T., Kenworthy, J., 2004. Trends in vehicle kilometres of travel in world cities, 1960–1990: underlying drivers and policy responses. Transp. Policy 11 (3), 287–298, http://doi.org/10.1016/j.tranpol.2004.01.002.

Cervero, R., Golub, A., 2007. Informal transport: a global perspective. Transp. Policy 14 (6), 445–457, http://doi.org/10.1016/j.tranpol.2007.04.011.

Chandran, A., Sousa, T.R.V., Guo, Y., Bishai, D., Pechansky, F., 2012. The vida no transito evaluation tea. (2012). Road traffic deaths in brazil: rising trends in pedestrian and motorcycle occupant deaths. Traffic Inj. Prev. 13 (Suppl. 1), S11–S16, http://doi.org/10.1080/15389588.2011.633289.

Chang, H.-L., Wu, S.-C., 2008. Exploring the vehicle dependence behind mode choice: Evidence of motorcycle dependence in Taipei. Transp. Res. A: Policy Pract. 42 (2), 307–320, http://doi.org/10.1016/j.tra.2007.10.005.

Chen, Chen, Lai, W.-T., 2011. The effects of rational and habitual factors on mode choice behaviors in a motorcycle-dependent region: evidence from Taiwan. Transp. Policy 18 (5), 711–718, http://doi.org/10.1016/j.tranpol.2011.01.006.

Clifton, K.J., Handy, S.L., 2001. Qualitative methods in travel behaviour research. Institute of Transportation Studies. University Of California, Davis (Retrieved from) 〈http://www.researchgate.net/profile/Kelly_Clifton/publication/228811651_Qualitative_methods_in_travel_behaviour_research/links/00b7d52c1baafa6ff0000000.pdf〉.

Companhia de Engenaria de Trânsito CET. (n.d.). Fatos e Estatísticas de Accidentes de Trânsito em São Paulo – 2011. São Paulo. Retrieved from 〈www.cetsp.com.br/media/186829/fat%20e%20est%202011.pdf〉.

Consejo Nacional, del Empleo, La Productividad, y el Salario Mínimo, Vital Y Móvil, 2015, July 24. Resolución 4/2015 - Salario Mínimo, Vital y Móvil. Modificación. Resolution. Retrieved from 〈https://www.boletinoficial.gob.ar/#!DetalleNormaBusquedaAvanzada/11487301/null〉.

Consórcio Motos, 2015, August 11. Honda CG 150 Titan 2015 a partir de R$ 147,86 mensais. Website of motorcycle dealership, Brazil. Retrieved from 〈https://www.consorciodemotos.com.br/noticias/honda-cg-150-titan-2015-a-partir-de-r-147–86-mensais〉.

Corporación Andina de Fomento, 2010. Observatorio de Movilidad Urbana para América Latina. Corporación Andina de Fomento: Caracas.

Creswell, J.W., 2012. Qualitative inquiry and research design: Choosing among five approaches. Sage. Retrieved from 〈https://books.google.com/books?hl=en&lr=&id=Ykruxor10cYC&oi=fnd&pg=PR1&dq=Cresswell+research+methods.&ots=4bl7kHq6Hj&sig=NT3YEdvjyFyCLdExgaUq2Nkv-Rg〉.

Dargay, J., Gately, D., Sommer, M., 2007. Vehicle ownership and income growth, worldwide: 1960–2030. Energy J., 143–170.

Davidson, J., 2006. NON-PROBABILITY (NON-RANDOM) SAMPLING. In V. Jupp, The SAGE Dictionary of Social Research Methods. 1 Oliver's Yard, 55 City Road, London England EC1Y 1SP United Kingdom: SAGE Publications, Ltd. Retrieved from 〈http://srmo.sagepub.com/view/the-sage-dictionary-of-social-research-methods/n130.xml〉.

De Sica, V., 1972. *The bicycle thief*. Image Entertainment, Incorporated.

Diaz Olvera, L., Plat, D., Pochet, P., Maïdadi, S., 2012. Motorbike taxis in the" transport crisis" of West and Central African cities. *EchoGéo*, (20). Retrieved from 〈http://%20echogeo.revues.org/13080〉.

Duffy, M., Robinson, T., 2004. An econometric analysis of motorcycle ownership in the UK. Int. J. Transp. Manag. 2 (3–4), 111–121, http://doi.org/10.1016/j.ijtm.2005.04.002.

Estache, A., Gómez-Lobo, A., 2005. Limits to competition in urban bus services in developing countries. Transp. Rev. 25 (2), 139–158.

Estupiñan, N., Santana, Manuel, Palacios, A., Rodríguez, Daniel A., 2012. Motorcycle Ownership and Use: The case of Latin America. Presented at the Conference on Advanced Systems for Public Transport, Santiago de Chile. Retrieved from 〈http://www.academia.edu/3856834/Motorcycle_ownership_and_use_The_case_of_Latin_America_Nicolas_Estupinan_et_al〉.

Fasakin, J.O., 2001. Some factors affecting daily profits of commercial motorcycles in Akure, Nigeria. Transp. Policy 8 (1), 63–69.

Freeman, L., 2011. There Goes the Hood: Views of Gentrification from the Ground Up. Temple University Press (Retrieved from) 〈https://books.google.com/books?hl=en&lr=&id=7GYL-g6_lnUC&oi=fnd&pg=PR5&dq=lance+freeman&ots=e1NGyEDObm&sig=dwnfOHgSn6lkfjkpt-uacsQy52Q〉.

Granovetter, M., 1985. Economic action and social structure: the problem of embeddedness. Am. J. Sociol., 481–510.

Guézéré, A., 2015. The reconstruction of shared taxis as rural transport due to the competition of motor bike taxis in Togo secondary cities. Case Stud. Transp. Policy 3 (2), 253–263, http://doi.org/10.1016/j.cstp.2014.10.003.

Gwilliam, K., 2003. Urban transport in developing countries. Transp. Rev. 23 (2), 197–216.

Haddon Jr, W., 1980. Advances in the epidemiology of injuries as a basis for public policy. Public Health Rep. 95 (5), 411.

Haworth, N., 2012. Powered two wheelers in a changing world—Challenges and opportunities. Accid. Anal. Prev. 44 (1), 12–18, http://doi.org/10.1016/j.aap.2010.10.031.

Hernández Sampieri, R., Fernández Collado, C., Baptista Lucio, P., 2010. Metodología de la Investigación (Quinta edición ed.).(J. Mares Chacón, Ed.) Mexico, México DF. McGraw-Hill/Interamericana Editores, SA de CV.

Hidalgo, D., Huizenga, C., 2013. Implementation of sustainable urban transport in Latin America. Res. Transp. Econ. 40 (1), 66–77, http://doi.org/10.1016/j.retrec.2012.06.034.

Huth, V., Füssl, E., Risser, R., 2014. Motorcycle riders' perceptions, attitudes and strategies: Findings from a focus group study. Transp. Res. F: Traffic Psychol. Behav. 25, 74–85, http://doi.org/10.1016/j.trf.2014.05.004.

Instituto Brasileiro de Geografia e Estatística. (n.d.). Pesquisa Mensal de Emprego - Estimativas para o Mês de Setembro de 2015 - Regiões Metropolitanas de: Recife, Salvador, Belo Horizonte, Rio De Janeiro, São Paulo E Porto Alegre. Retrieved from ⟨ftp://ftp.ibge.gov.br/Trabalho_e_Rendimento/Pesquisa_Mensal_de_Emprego/fasciculo_indicadores_ibge/2015/pme_201509pubCompleta.pdf⟩.

Kerlinger, F.N., Lee, H.B., Ayala, L.E.P., Magaña, I.M., 2002. Investigación del comportamiento. McGraw-Hill México City, México. Retrieved from ⟨http://www.sidalc.net/cgi-bin/wxis.exe/?IsisScript=AGRIUAN.xis&method=post&formato=2&cantidad=1&expresion=mfn=027562⟩.

Kopp, P., 2011. The unpredicted rise of motorcycles: a cost benefit analysis. Transp. Policy 18 (4), 613–622, http://doi.org/10.1016/j.tranpol.2011.03.002.

Kumar, A., 2011. Understanding the Emerging Role of Motorcycles in African Cities: A Political Economy Perspective. Retrieved from ⟨https://www.openknowledge.worldbank.com/handle/10986/17804⟩.

Li, M.-D., Doong, J.-L., Huang, W.-S., Lai, C.-H., Jeng, M.-C., 2009. Survival hazards of road environment factors between motor-vehicles and motorcycles. Accid. Anal. Prev. 41 (5), 938–947, http://doi.org/10.1016/j.aap.2009.05.009.

Lin, M.-R., Kraus, J.F., 2009. A review of risk factors and patterns of motorcycle injuries. Accid. Anal. Prev. 41 (4), 710–722, http://doi.org/10.1016/j.aap.2009.03.010.

Lizarraga, C., 2012. Expansión metropolitana y movilidad: el caso de Caracas. EURE (Santiago) 38 (113), 99–125.

Metrô São Paulo. 2008, December. PESQUISA ORIGEM - DESTINO /2007 – Relatório Síntese das Informações – Pesquisa Domiciliar. Retrieved from ⟨http://www.stm.sp.gov.br/images/stories/Pitus/Pitu2025/Pdf/sintese_od_2007.pdf⟩.

Ministerio del Trabajo. (n.d.). Salario mínimo legal en Colombia. Retrieved from ⟨http://obiee.banrep.gov.co/analytics/saw.dll?Go&Path=/shared/Consulta%20Series%20Estadisticas%20desde%20Excel/1.%20Salarios/1.1%20Salario%20minimo%20legal%20en%20Colombia/1.1.1%20Serie%20historica&Options=rdf&NQUser=salarios&NQPassword=salarios&lang=es⟩.

Moustakas, C., 1994. Phenomenological research methods. Sage Publications. Retrieved from ⟨https://books.google.com/books?hl=en&lr=&id=pp11AwAAQBAJ&oi=fnd&pg=PP1&dq=Moustakas,+C.+(1994)+Phenomenological+research+methods.&ots=O7QbWtLAG1&sig=R_uETy3ETFWsyB6GkZZj8hH_qGQ⟩.

Musselwhite, C.B.A., Avineri, E., Susilo, Y.O., Bhattachary, D., 2012. Public attitudes towards motorcyclists' safety: a qualitative study from the United Kingdom. Accid. Anal. Prev. 49, 105–113, http://doi.org/10.1016/j.aap.2011.06.005.

Observatório Metropolitano. (n.d.). Area Metropolitana de Barranquilla. Website of the authority of the Metropolitan Area of Barranquilla. Retrieved from ⟨http://www.ambq.gov.co/secciones.php?seccion=6ea9ab1baa0efb9e19094440c317e21b&topmenu=Y⟩.

Pardo, C.F., Hagen, J.X., Burbano Valente, J., 2012. Entrevistas sobre motociclistas en América Latina – informe final adenda. Unpublished manuscript.

Presidência da República, 2014, December 29. Decreto no. 8.381, de 29 de dezembro de 2014. Decree. Retrieved from ⟨http://www.planalto.gov.br/ccivil_03/_ato2011–2014/2014/decreto/d8381.htm⟩.

Ratanavaraha, V., Jomnonkwao, S., 2013. Community participation and behavioral changes of helmet use in Thailand. Transp. Policy 25, 111–118, http://doi.org/10.1016/j.tranpol.2012.11.002.

Redman, L., Friman, M., Gärling, T., Hartig, T., 2013. Quality attributes of public transport that attract car users: a research review. Transp. Policy 25, 119–127, http://doi.org/10.1016/j.tranpol.2012.11.005.

Reolom, M., Italiani, R., 2014, May 24. SP tem maior lentidão da história: 344 km - São Paulo. O Estado de São Paulo. Retrieved from ⟨http://sao-paulo.estadao.com.br/noticias/geral,sp-tem-maior-lentidao-da-historia-344-km-imp-,1170914⟩.

Rodríguez, D.A., Santana, M., Pardo, C.F., 2015. La motocicleta en América Latina: caracterización de su uso e impactos en la movilidad en cinco ciudades de la región. Retrieved from ⟨http://scioteca.caf.com/handle/123456789/754⟩.

Rogers, N., 2008. Trends in Motorcycles Fleet Worldwide. Presentation to Joint OECD/ITF Transport Research Committee Workshop on Motorcycling Safety.

Sánchez Jabba, A., 2011. La economía del mototaxismo: El caso de Sincelejo. BANCO DE LA REPÚBLICA-ECONOMÍA REGIONAL. Retrieved from ⟨http://econpapers.repec.org/paper/col000102/008200.htm⟩.

Santos, G., Behrendt, H., Maconi, L., Shirvani, T., Teytelboym, A., 2010. Part I: externalities and economic policies in road transport. Res. Transp. Econ. 28 (1), 2–45, http://doi.org/10.1016/j.retrec.2009.11.002.

Shoup, D.C., 2005. The high cost of free parking (Vol. 206). Planners Press Chicago. Retrieved from ⟨https://server1.tepper.cmu.edu/CMUPark/The-High-Cost-of-Free-Parking.pdf⟩.

Silva, E.R. da, Cardoso, B.C., Santos, M.P. de S., 2011. O aumento da taxa de motorização de motocicletas no Brasil. Revista Brasileira de Administração Científica, 2(2). http://dx.doi.org/10.6008/ESS2179-684X.2011.002.0004.

Simons, D., Clarys, P., De Bourdeaudhuij, I., de Geus, B., Vandelanotte, C., Deforche, B., 2014. Why do young adults choose different transport modes? A focus group study. Transp. Policy 36, 151–159, http://doi.org/10.1016/j.tranpol.2014.08.009.

Small, K.A., Verhoef, E.T., 2007. The Economics of Urban Transportation. Routledge, New York.

Thomson, I., Bull, A., 2001. La congestión del tránsito urbano: causas y consecuencias económicas y sociales. Santiago de Chile: Naciones Unidas, Comisión Económica para América Latina y el Caribe, División de Recursos Naturales e Infraestructura, Unidad de Transporte.

Thomson, I., Bull, A., 2002. La congestión del. Rev De La CEPAL 76, 109.

Tunnicliff, D., Watson, B., White, K.M., Lewis, I., Wishart, D., 2011. The social context of motorcycle riding and the key determinants influencing rider behavior: a qualitative investigation. Traffic Inj. Prev. 12 (4), 363–376.

Uno Motos. (n.d.). Linea calle: Precios Actualizados. Website of motorcycle dealership. Retrieved from ⟨http://www.unomotos.com.ar/categoria_calle⟩.

Van Manen, M., 1990. Researching lived experience: Human science for an action sensitive pedagogy. Suny Press (Retrieved from) ⟨https://books.google.com/books?hl=en&lr=&id=ub9GAAAAQBAJ&oi=fnd&pg=PR7&dq=Van+Manen,+M.+(1990)+Researching+lived+experience:+Human+science+for+an+action+sensitive+pedagogy&ots=F_yo_qcKap&sig=I9BHZAlUXGwRfu8SbjoQbOut9lI⟩.

Vasconcellos, E.A.D., 2005. Urban change, mobility and transport in São Paulo: three decades, three cities. Transp. Policy 12 (2), 91–104, http://doi.org/10.1016/j.tranpol.2004.12.001.

Vasconcellos, E.A.D., 2013. Road safety impacts of the motorcycle in Brazil. Int. J. Inj. Control Saf. Promot. 20 (2), 144–151.

Vasconcellos, E.A., 2008. O custo social da motocicleta no Brasil. Rev. Dos. Transp. Públicos–ANTP Ano 30, 31.

Vasconcellos, E.A., 2014. Urban Transport Environment and Equity: The Case for Developing Countries. Routledge (Retrieved from) ⟨https://books.google.com/books?hl=en&lr=&id=O2B9AwAAQBAJ&oi=fnd&pg=PP1&dq=vasconcellos+2014&ots=qMS9imEiZ-&sig=zUoq_VCQ1cM6WJ4oE-LWpZUbsRI⟩.

World Health Organization, 2013. WHO Global Status Report on Road Safety 2013: Supporting a Decade of Action. World Health Organization (Retrieved from) ⟨http://apps.who.int/iris/handle/10665/78256⟩.

Yamamoto, T., 2009. Comparative analysis of household car, motorcycle and bicycle ownership between Osaka metropolitan area, Japan and Kuala Lumpur, Malaysia. Transportation 36 (3), 351–366, http://doi.org/10.1007/s11116–009–9196-x.